Angewandte Chemie
Eine Zeitschrift der Gesellschaft Deutscher Chemiker

Supporting Information

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Giant *ab*-Plane Birefringence in Quasi-1D Fibrous Red Phosphorus

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Abstract: Quasi-one-dimensional (quasi-1D) van der Waals crystal fibrous red phosphorus (RP) exhibits pronounced in-plane optical anisotropy, positioning it as a potential candidate for polarization-related micro-nano devices. Unfortunately, a comprehensive investigation into the complex refractive index of fibrous RP and the structure-activity relationship connecting the distinctive quasi-1D structure with optical anisotropy is currently deficient. Herein, we have collectively determined the complex refractive index of the fibrous RP flakes within the ab-plane through Kramers–Kronig (KK) analysis and theoretical calculation. Notably, the maximum birefringence of fibrous RP reaches 0.642@475 nm with an absolute extinction coefficient of only 0.08, superior to the reported traditional optical crystals and the emerging low-dimensional materials as well. The remarkable birefringence can be attributed to the synergistic influence of the large electronic dipole polarizability, anisotropic electron density distribution and the distortion of stereochemically active lone pair (SCALP). This work demonstrates the potential of fibrous RP for polarization-sensitive devices, illuminating possibilities to exploit novel giant birefringent crystals based on the structure-activity relationship.

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Materials Synthesis

Growth of the fibrous RP flakes

Fibrous RP flakes were obtained via a CVT method as previously reported[31]. Typically, amorphous RP (Alfa Aesar, 99.999%, ~400 mg), Sn powder (Alfa Aesa, 99.99%, ~20 mg), I_2 (Alfa Aesa, 99.9%, ~15 mg) were mixed and loaded into a quartz ampule at one end. The SiO_2/Si substrate was put at the other end of the quartz ampule. Then the ampoule was evacuated to 1.0×10^{-3} Pa and sealed (MRVS-1002, Partulab). After that, the sealed ampule was loaded into a two-zone furnace and heated according to the temperature program. Specifically, the furnace temperature was raised to 620 °C for the growth zone (610 °C for the source zone) at a rate of 4 °C/min, maintained at this temperature for 2 hours, and then gradually cooled down to room temperature overnight.

Preparation of samples for TEM testing

In a standard liquid exfoliation experiment, add the as-obtained SiO_2/Si loaded with fibrous RP flakes into a bottle containing 5 mL Isopropanol (IPA). To prevent oxidation, the bottle was filled with argon, tightly sealed with parafilm, and subjected to ultrasonication using a bath ultrasonic power of 100 W (KQ2200DB, Kunshan Ultrasonic Instruments) for 60 min. After exfoliation, the suspensions were centrifuged at 3000 rpm for 5 min to remove the residue. Take a small amount of centrifuged supernatant and drop it onto a copper mesh to obtain a sample for TEM testing.

Sample Characterizations

Optical micrograph

The optical images were acquired using a microscope equipped with optical objectives of 50x and 100x magnification (BX 51M, Olympus).

Raman spectrum and atomic force microscope (AFM)

Raman spectra were acquired using a laser scanning confocal microscope (Alpha300RAS, Witec) with 532 nm laser excitation at room temperature. Collection time was 15 seconds with 4 accumulations, employing a laser power of approximately 300 μW. The thickness of as-obtained fibrous RP flakes on the SiO_2/Si was examined with AFM in AC mode (Alpha300RAS, Witec).

X-ray diffraction (XRD)

The powder X-ray diffraction (XRD) patterns were obtained using a Bruker D8 Advance diffractometer equipped with Cu-Kα radiation (λ=1.5406 Å) at 40 kV and 40 mA.

Transmission electron microscopy (TEM) and high-resolution TEM

The low-magnification TEM images were obtained using H-7650B at 80 kV. The high angle annular dark field-scanning transmission electron microscopy image and the lattice fringe images of fibrous RP were recorded by high-resolution TEM (JEOL JEM 2100F) equipped with an energy dispersive X-ray spectroscope (EDX).

Polarization-resolved reflection spectrum and mappings

The polarization-resolved optical images were captured by rotating the sample by a multifunctional spectral imaging microscope (Alpha300RAS, Witec) equipped with a 100 times magnification optical objective. The polarization-resolved reflection spectra and reflectance spectra were collected using polarized white light excitation at room temperature. Collection time for every angle was 0.1 seconds with 4 accumulations. Corresponding polarization-resolved contrast spectra were calculated by Equation 3. The reflection spectra mappings were taken with a step of 0.3 μm.

Theoretical calculations

The Vienna *ab* initio simulation package (VASP) was employed for conducting first-principles calculations based on density functional theory and projector augmented wave (PAW) method[1,2]. The exchange-correlation potential was treated with the generalized gradient approximation (GGA) of the Perdew-Burke-Ernzerhof (PBE) standard functional[3]. The plane-wave cutoff energy was set at 520 eV. The DFT-D3 method was employed to account for van der Waals interactions in all calculations[4]. The structural optimization initiates with the atomic arrangement in bulk fiber red phosphorus, during which the lattice constant and atomic positions are completely relaxed. Additionally, in the structural optimization of few-layer fibrous RP, atoms were allowed to relax thoroughly within 2D slab models comprising one to four layers. A vacuum layer of 20 Å was incorporated to prevent unnecessary interactions between periodic replicas. The total energy and force acting on each atom were converged to 10^{-5} eV and 0.03 eV $Å^{-1}$, respectively. The 4×4×7 and 2×2×1 gamma-centered k-point grids were employed to calculate the bulk fibrous RP and few-layer fibrous RP along the *c*-axis, respectively. The calculation of the dielectric function was carried out to determine the refractive index and extinction coefficient. In the optical property calculation, more refined gamma-centered k-point grids of 6×6×11 and 5×5×1 were applied in the calculation of bulk fibrous RP and few-layer fibrous RP, respectively. Moreover, the more precise energy tolerance of 10^{-7} eV was utilized to ensure computational accuracy. Described by the real (e_1) and imaginary (e_2) parts, the dielectric function allows the derivation of the refractive index (n) and extinction coefficient (κ) through the following equation:

$$\begin{cases} n = \left(\dfrac{\sqrt{e_1^2+e_2^2}+e_1}{2}\right)^{1/2} \\ \kappa = \left(\dfrac{\sqrt{e_1^2+e_2^2}-e_1}{2}\right)^{1/2} \end{cases}. \quad (1)$$

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Results and Discussion

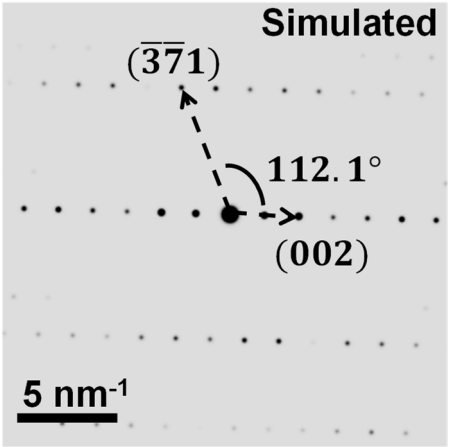

Figure S1 The simulated electron diffraction pattern of fibrous red phosphorus (RP) along the zone axis [$\bar{7}30$] using Crystal Maker software.

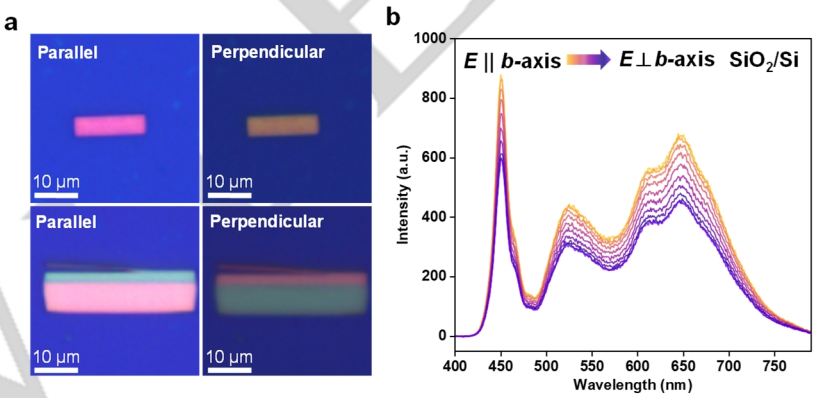

Figure S2 a) Optical micrographs depicting fibrous RP flakes illuminated with horizontal and vertical linearly polarized white light at the same power level. b) Polarization-resolved reflection spectra of SiO_2/Si substrate. During the testing process, maintain the position of the sample unchanged while modulating the polarization direction of the incident white light from horizontal to vertical using a half-wave plate.

For isotropic substrates, the reflection spectra demonstrate polarization-dependent intensity variations. This is attributed to the fact that while linearly polarized white light undergoes a change in its polarization direction through a half-wave plate at a consistent power, the resulting intensity loss varies.

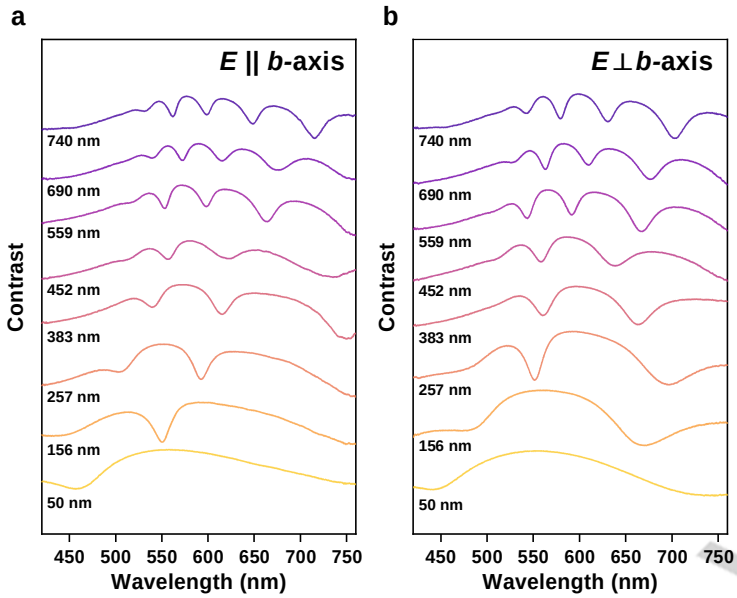

Figure S3 Thickness-dependent contrast spectra of the fibrous RP flakes under a) parallel (||) and b) perpendicular (⊥) configurations, respectively.

Specifically, for a four-layer medium (air/fibrous RP/SiO_2/Si) system, the reflected intensity by the fibrous RP flakes and the substrate can be expressed as follows:

$$I_{FRP} = I_0 \left| \frac{r_{01}e^{i(\phi_1+\phi_2)} + r_{12}e^{-i(\phi_1-\phi_2)} + r_{23}e^{-i(\phi_1+\phi_2)} + r_{01}r_{12}r_{23}e^{i(\phi_1-\phi_2)}}{e^{i(\phi_1+\phi_2)} + r_{01}r_{12}e^{-i(\phi_1-\phi_2)} + r_{01}r_{23}e^{-i(\phi_1+\phi_2)} + r_{12}r_{23}e^{i(\phi_1-\phi_2)}} \right|^2 \quad (2)$$

$$I_{Substrate} = I_0 \left| \frac{r_{02} + r_{23}e^{-2i\phi_2}}{1 + r_{02}r_{23}e^{-2i\phi_2}} \right|^2 \quad (3)$$

where $r_{ij} = (\tilde{n}_i - \tilde{n}_j)/(\tilde{n}_i + \tilde{n}_j)$ and $\phi_i = 2\pi\tilde{n}_i d_i/\lambda$ represent reflection coefficients and phase shift, respectively. In detail, $\tilde{n}_i$, $\tilde{n}_j$ represent the complex refractive indices of medium layer, d_i denotes the thickness of the corresponding medium layer, and λ is the wavelength.

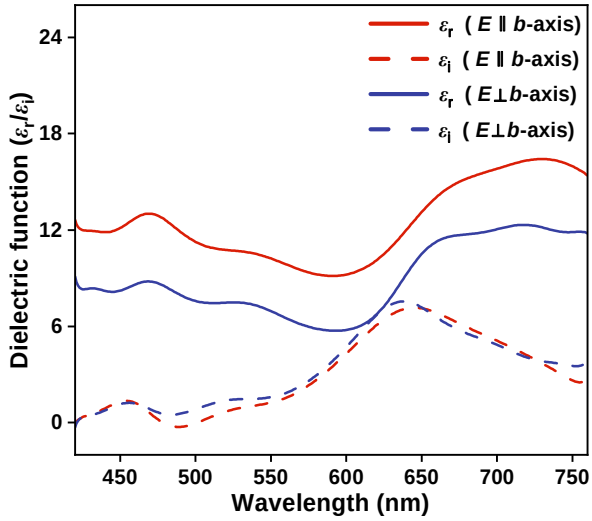

Figure S4 The real (ε_r) and imaginary (ε_i) parts of dielectric function for the fibrous RP flake along the direction parallel or perpendicular to the*b*-axis.

After extracting the Refractive index (n) and extinction coefficient (κ) of the fibrous RP flakes through Kramers–Kronig (KK) relations, the real (ε_r) and imaginary (ε_i) parts of dielectric function can be further derived by the following equation:

$$\varepsilon_r(\omega) = n^2 - \kappa^2, \tag{4}$$

$$\varepsilon_i(\omega) = 2n\kappa. \tag{5}$$

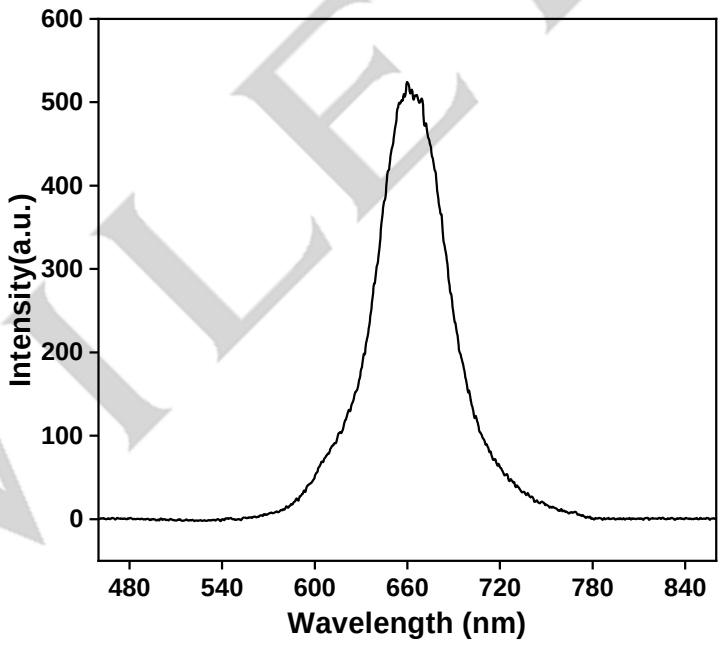

Figure S5 Photoluminescence (PL) spectrum of fibrous RP flakes. The peak is located at approximately 660 nm.

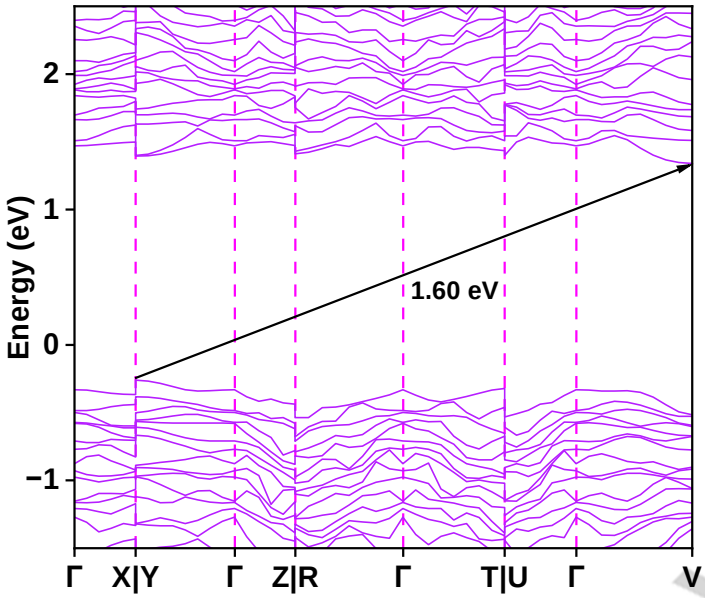

Figure S6 The electronic structure of the bulk fibrous RP. The difference between the calculated band gap and experimental results is attributed to the intrinsic underestimation associated with the GGA-PBE functional.

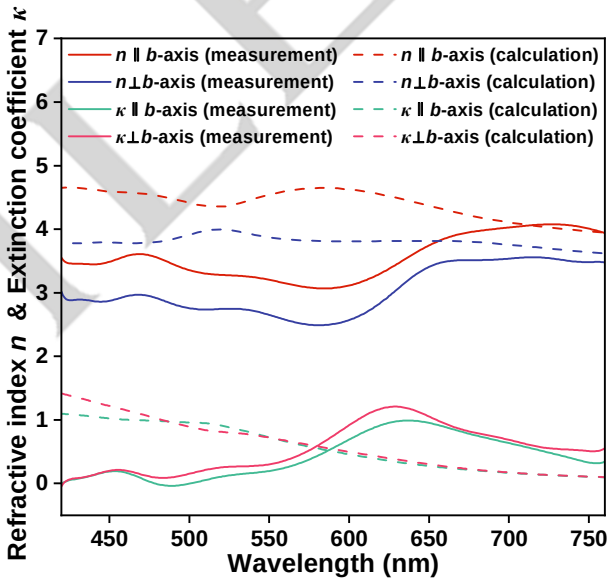

Figure S7 Comparison of refractive index (n) and extinction coefficient (κ) obtained by theoretical calculation and Kramers–Kronig relations. Due to the existence of exciton effects in actual optical processes, the difference between theoretical calculation results and experimental results is common.

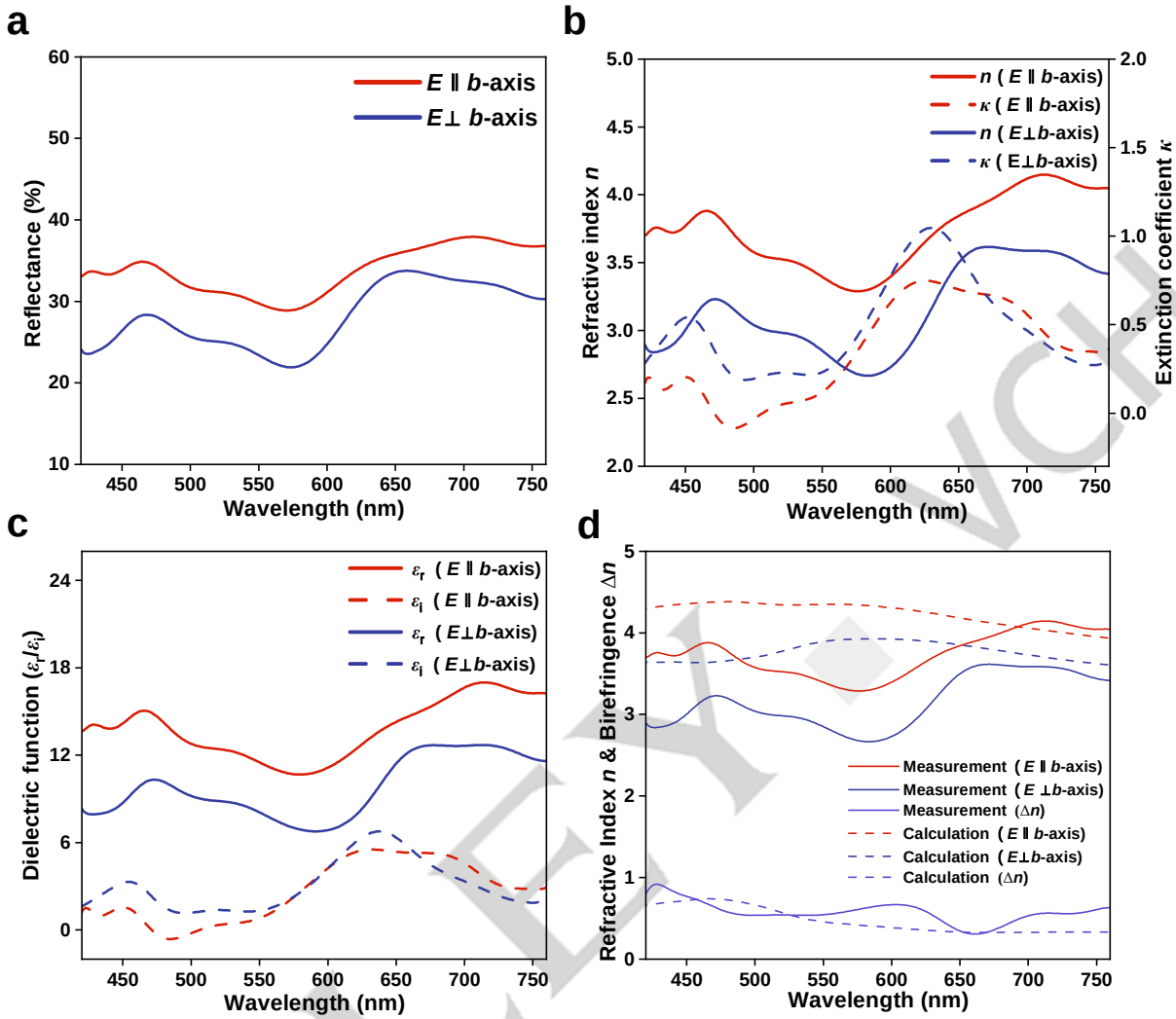

Figure S8 a) Reflectance spectrum, b) refractive index (n) and extinction coefficient (κ), and c) real (ε_r) and imaginary (ε_i) parts of dielectric function of the sample 1 along the direction parallel or perpendicular to the *b*-axis of fibrous RP. d) Comparison of the refractive index and birefringence of the sample 1 obtained by measurement and calculation.

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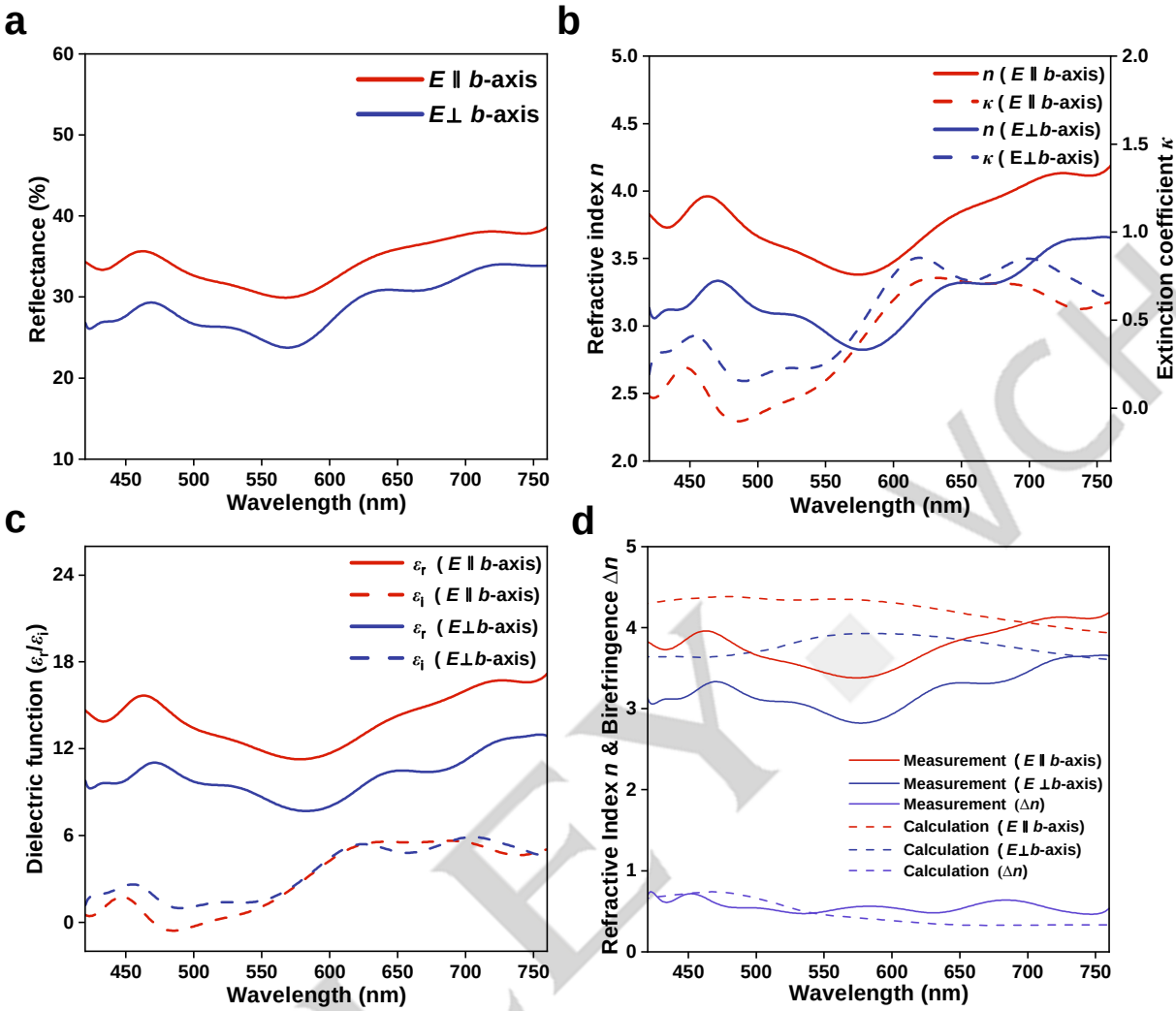

Figure S9 a) Reflectance spectrum, b) refractive index (n) and extinction coefficient (κ), and c) real (ε_r) and imaginary (ε_i) parts of dielectric function of the sample 2 along the direction parallel or perpendicular to the *b*-axis of fibrous RP. d) Comparison of the refractive index and birefringence of the sample 2 obtained by measurement and calculation.

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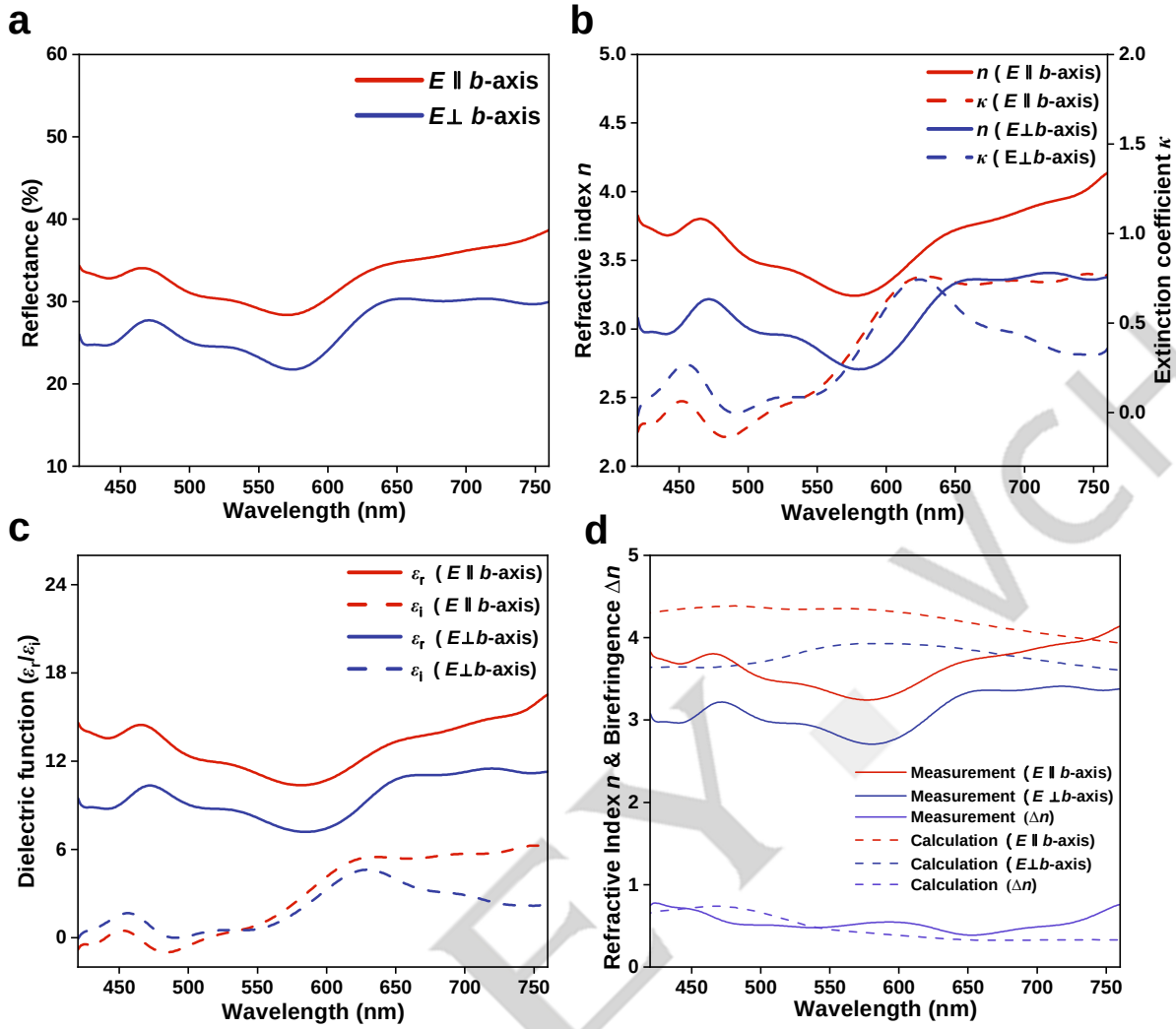

Figure S10 a) Reflectance spectrum, b) refractive index (n) and extinction coefficient (κ), and c) real (ε_r) and imaginary (ε_i) parts of dielectric function of the sample 3 along the direction parallel or perpendicular to the *b*-axis of fibrous RP. d) Comparison of the refractive index and birefringence of the sample 3 obtained by measurement and calculation.

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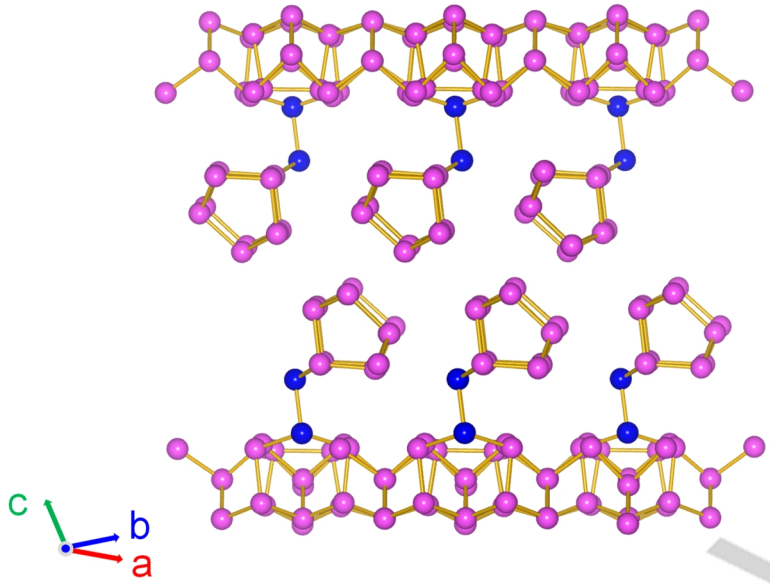

Figure S11 The typical crystal structure of adjacent pentagonal phosphorus tubes arranged vertically in violet phosphorus.

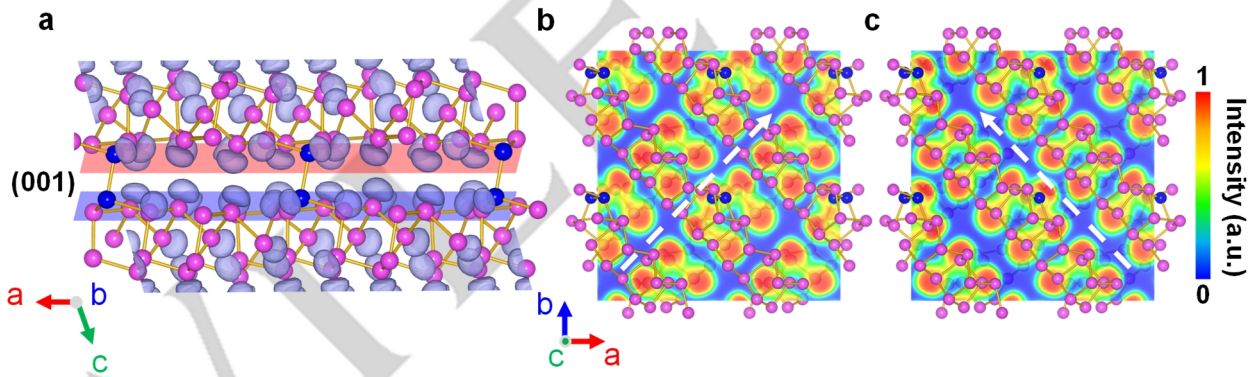

Figure S12 a) Three-dimensional visualization of the electron localization function (ELF) with an isosurface value of 0.9 for violet phosphorus. b), c) Two-dimensional ELF slices of violet phosphorus intersecting along the [001] direction, cutting through the blue and red planes shown in Figure S12a, respectively. The white arrow indicates the extension direction of the pentagonal phosphorus tube.

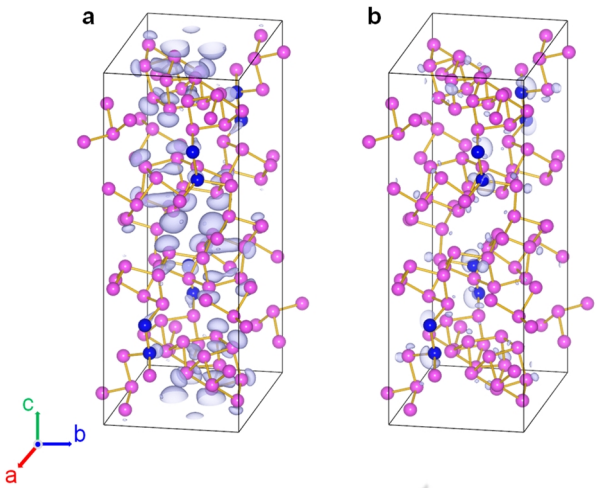

Figure S13 a), b) Highest Occupied Crystal Orbital (HOCO) and Lowest Unoccupied Crystal Orbital (LUCO) of violet phosphorus.

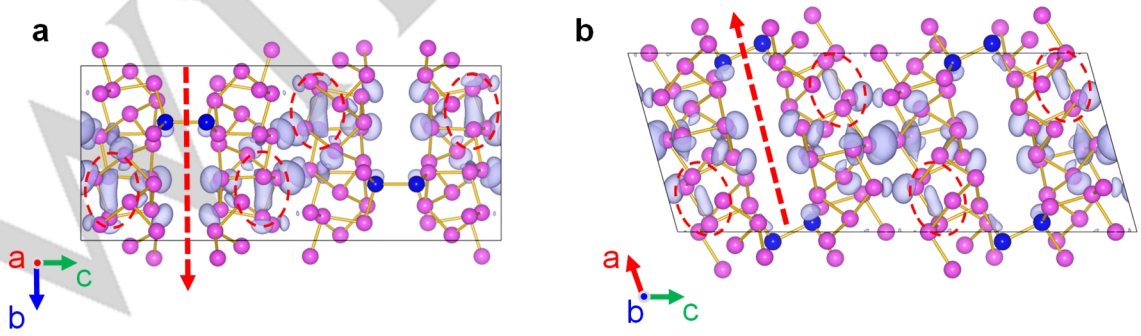

Figure S14 HOCO of violet phosphorus viewed from a) [100] and b) [010] directions. The red dash circles highlight the delocalized electron and the red dash line indicates the direction of electron delocalization.

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Author contributions

W. Chen synthesized and characterized the samples, conducted the optical measurement and data analysis, and performed the DFT calculations. W. Chen wrote the original draft. B. Zhang and K. Tao gave the guidance about numerical analysis and theoretical calculation. Q. Li and J.-L. Sun provided experimental setup and guidance on optical characterizations. Q. Yan conceived and supervised the project. All authors participated in discussion and editing of the manuscript.

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